Tecnológico de Monterrey | Graduate School

Master in Applied Artificial Intelligence (MNA)

“Multi-Agent Reinforcement Learning for Collaborative Box-Pushing”

Tsinghua University

Presented by:

- Julia Gabriela Pinedo

Project Advisor: Dr. Guillermo Mota Medina

Stakeholder: Dr. Fuxi Wen

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Capstone Project Presentation

Problem Statement

The **box-pushing problem** is a crucial benchmark in cooperative robotics, as it models scenarios where one or more robots must push a heavy box toward a goal region while avoiding obstacles and walls [1]. This demands superior coordination between robots, which has led to its adoption in **Multi-Agent Reinforcement Learning (MARL)** [2][3].

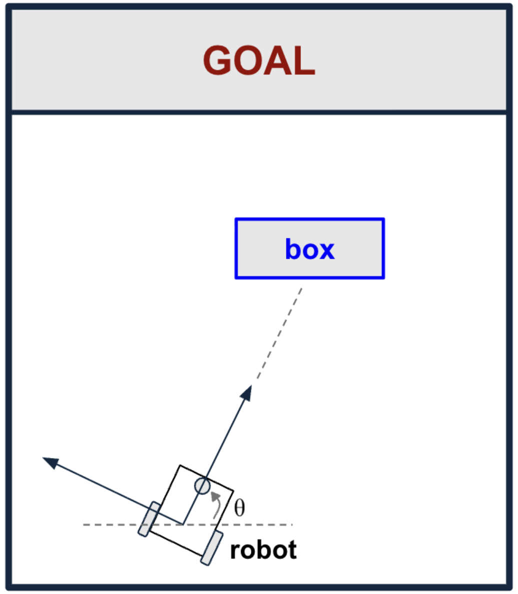
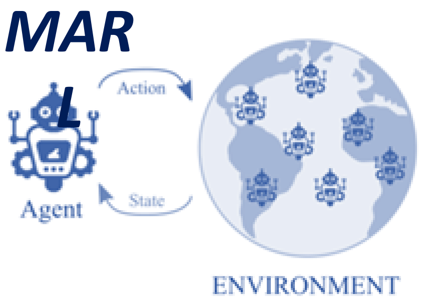


Figure 1. Diagram of a classic paradigm of the box-pushing problem.

However, some authors note that canonical MARL-based approaches underperform in comparison to Single-Agent Reinforcement Learning (SARL) solutions [2][3], motivating the need to improve collaborative learning in MARL systems by establishing a SARL baseline to highlight the current limitations and provide a clearer benchmarks for developing more robust multi-agent solutions.

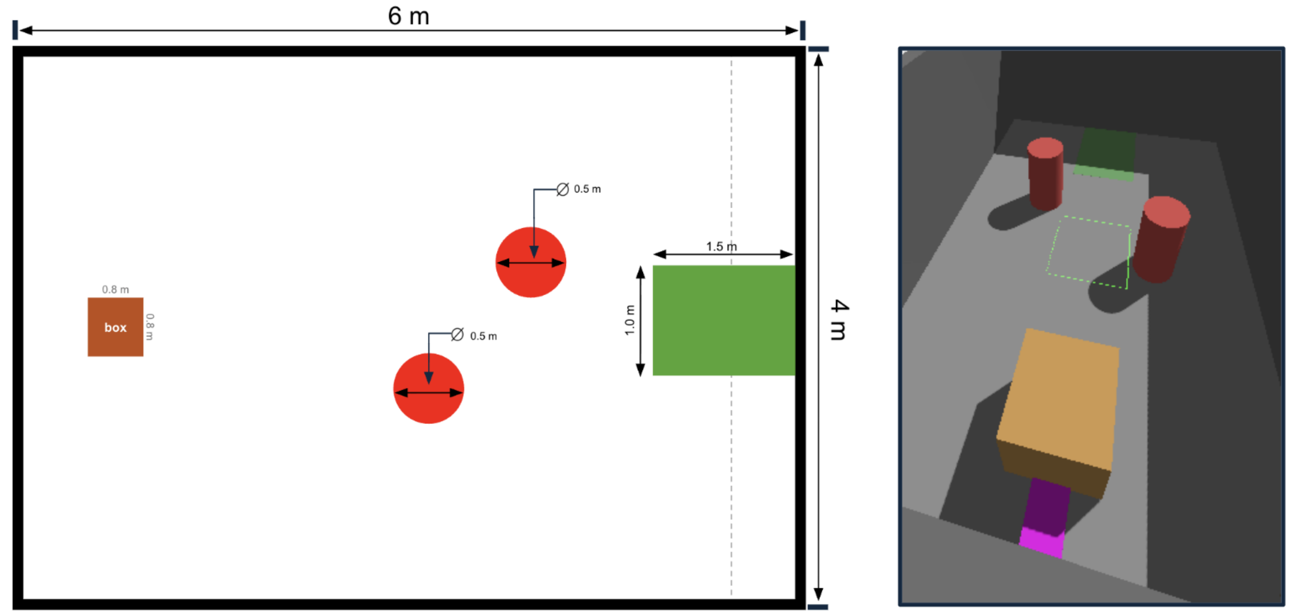


This project follows the CRISP-ML process, from data understanding and simulation design to modeling, evaluation, and deployment preparation. Each stage corresponds to the workflow shown in this presentation.

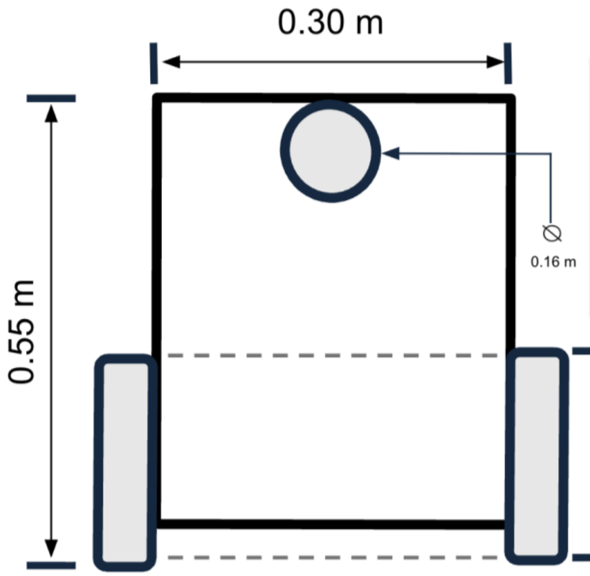
Problem Statement

Since the project is based on Reinforcement Learning techniques, we do not rely on a traditional static dataset. Instead, the data is generated through interaction with the simulation in Gazebo + ROS 2. The environment consists of:

Simulation arena, with static obstacles, and a pushable box



Custom diff-drive robot



Part	Dimensions
Base (main body)	Length: 0.55 m Width: 0.30 m Height: 0.12 m Mass: ~7.5 kg
Wheels (x2)	Radius: 0.08 m Width: 0.03 m Mass: 0.3 kg
Caster	Radius: 0.08 m Mass: 0.2 kg

Figure 2. Simulation environment for this project. Sensor readings are standardized and synchronized to ensure consistent temporal alignment and generate a dataset.

Key Findings and Insights

To adapt this project into a **CRISP-ML methodology**, a **manual** simulation of the environment was run. Each episode corresponds to a simulated full run of the diff-drive robot inside the Gazebo arena, capturing sensor readings, distances to obstacles and goal regions, and discrete event flags [7].

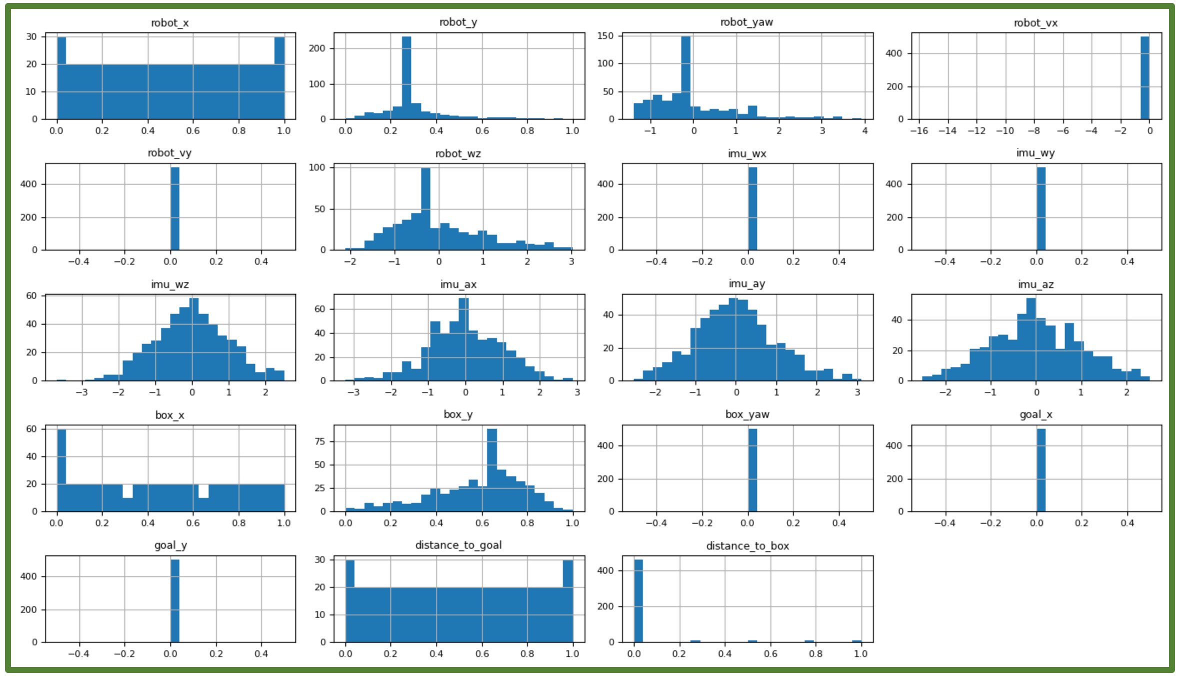


Figure 3. Histogram generated with the main continuous variables.

Type	Variables
Spatial and Orientation	robot_x, robot_y, robot_yaw, box_x, box_y, box_yaw, goal_x, goal_y
Kinematics	robot_vx, robot_vy, robot_wz, wheel_l_rad_s, wheel_r_rad_s
Inertial Features (IMU)	imu_wx, imu_wy, imu_wz, imu_ax, imu_ay, imu_az
Commands	cmd_vx, cmd_wz
Reward Components	r_progress, r_collision, r_efficiency, r_rot_pen, r_goal, r_total

Type	Variables
Episode Tracking	episode, step (integer)
Event Flags	contact_flag, in_goal, done (boolean)
Outcome Label	imu_ax, imu_ay, imu_az
Commands	done_reason (object/string)

Table 1-2. Description of the numerical and categorical variables.

Key Findings and Insights

The **correlation matrix** confirms physical consistency between position, kinematics, and sensor data. Overall, effective navigation arises from stable and progressive motion rather than abrupt acceleration.

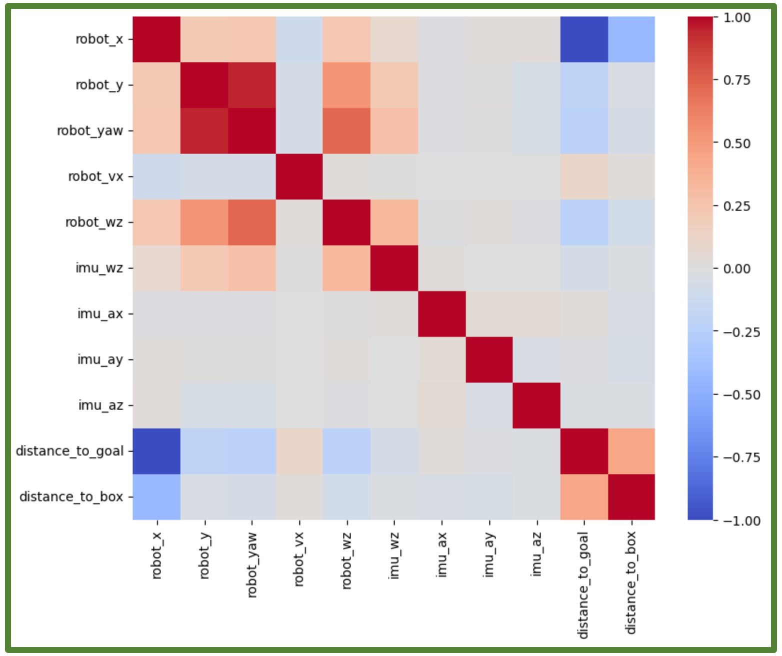


Figure 4. Correlation matrix.

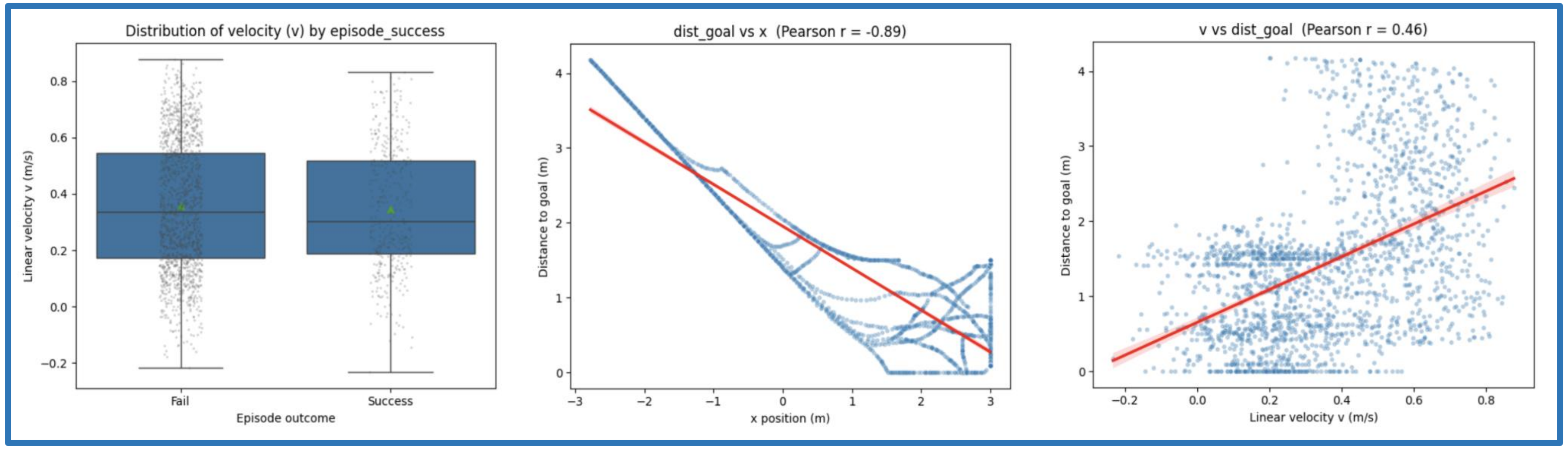


Figure 5. Bivariate relationships between key motion and task variables.

***Bivariate analysis** reveals that:*

- *Task success depends on consistent forward motion, not higher speed.*
- *x-position correlates inversely with goal distance, confirming progress direction.*
- *Velocity increases when far from target, showing adaptive speed behavior.*

Proposed Solution

To **identify the solution** that best aligns with the project objectives, the following core evaluation metrics were defined to assess the model performance [7][8][9][10]:

Metric
Eval success rate ≥ **0.20**
Median steps-to-success ≤ **80**
Positive learning trend on eval **(rolling/local slope > 0)**
No overfitting signal (eval_return − train_return at eval points**) ≥ −1.0**

Table 3. Floor criteria.

Evaluation success rate: *Measures task completion under evaluation conditions and remains robust to variations in reward shaping.*
Episode length/steps to success (efficiency): *Evaluates how quickly the robot completes successful episodes, with shorter trajectories indicating higher operational efficiency.*
Episode return over time (learning curve): *This measures whether the policy improves its ability to maximize cumulative rewards, reflecting how effectively the robot learns to push the box toward the goal as training progresses.*
Success rate (goal completion): *Captures the fraction of episodes in which the box reaches the goal region, serving as a direct and interpretable indicator of task achievement independent of reward shaping.*

Algorithm	Eval success rate	Median steps-to-success	Positive learning trend	No overfitting signal	Execution time (mins)
Monte Carlo Policy Gradient (REINFORCE)	0.35	62.5 steps	0.176	0.382	6.42
Deep Q-Network (DQN)	0.35	60 steps	0.083	1.335	6.37
Distributional DQN (C51)	0.35	73 steps	-0.002	-0.726	6.32
Soft Actor-Critic (SAC)	0.35	58 steps	-0.124	0.921	6.38
Advantage Actor-Critic (A2C)	0.76	57 steps	2.614	0.9502	6.11
Proximal Policy Optimization (PPO)	0.68	58 steps	2.444	0.9504	6.22

Proposed Solution

Based on the empirical results of the six candidate models, **Advantage Actor-Critic (A2C)** was selected as the final model. To further enhance model stability, a bagging technique was incorporated into the implementation.

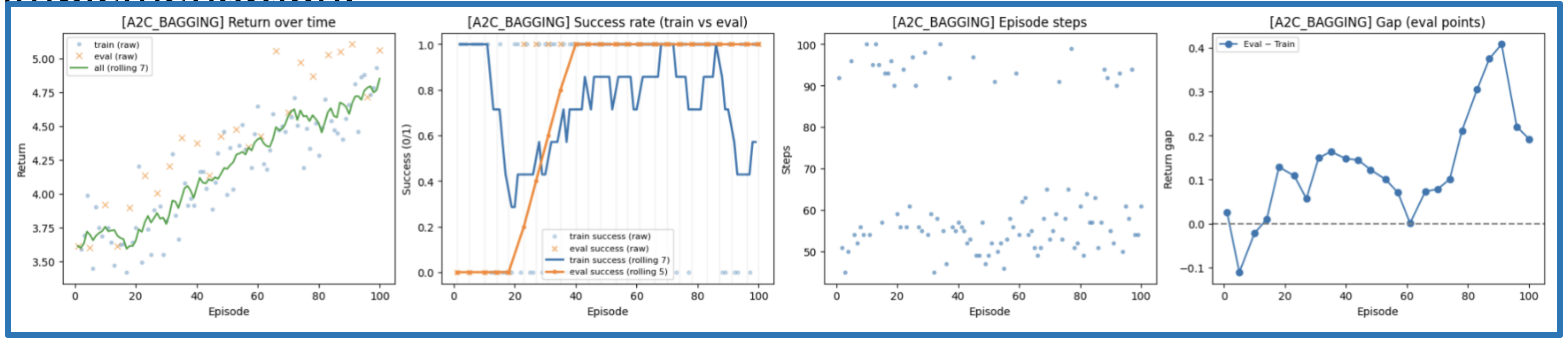


Figure 6. Graphs of the A2C Bagging model.

Algorithm	Eval success rate	Median steps-to-success	Positive learning trend	No overfitting signal	Execution time (mins)
Advantage Actor-Critic (A2C) + Bagging	0.79	56 steps	0.067	0.127	8.40

Table 5. Performance of the final model.

*These results satisfy the **following objectives:***

1. *Establish a Single-Agent Reinforcement Learning baseline.*
2. *Extend the same SARL baseline algorithm to a multi-agent setting.*
3. *Design and implement a robust Multi-Agent Reinforcement Learning framework that enables effective cooperation among multiple differential-drive robots in simulated environments.*
4. *Integrate a custom differential-drive robot model equipped with odometry, IMU (Inertial Measurement Unit), and contact detection sensors to support a realistic RL environment in Gazebo + ROS 2.*
5. *Evaluate performance across key metrics such as: success rate, efficiency, robustness under agent failures, and adaptability to obstacles.*

Key Business Recommendations

To ensure the solution's long-term scalability and reproducibility, the following implementation and environment recommendation were defined. These serve as the **core strategic guidelines** to strengthen the SARL foundation:

Model's Performance:

1. ***Stabilize the single-agent baseline.*** *Include stronger penalties where errors are most costly to encourage a smoother and more purposeful behavior.*

2. *Implement a tighter exploration-to-exploitation schedule, for example, by gradually reducing the entropy bonus to allow the agent to shift from broad exploration toward more consistent, goal-directed actions as training progresses.*

Model's Implementation:

1. ***Standardization of the computational environment.*** *Although the use of Windows Subsystem for Linux has supported initial experiments, migrating to a system with Ubuntu Linux 22.04 for the simulation is recommended for best compatibility.*
2. *Shipping a single Dockerfile (CPU + optional CUDA tag) will guarantee that the runs are reproducible on any machine.*
3. ***Automate and share the analysis pipeline.*** *To make the result visualization transparent and reusable, a Jupyter Notebook should be created to import metrics, generate the plots automatically, the summary table, and save the output figures to ingest them into reports or LaTeX documents for easy tracking*

Expected Benefits and Outcomes

Since this project remains in an **academic setting**, it incurs no significant financial costs beyond personal computing resources. The focus lies on the expected benefits derived from improved stability, reproducibility and scalability of the RL framework.

Quantifiable Benefits	***Qualitative Benefits***
Faster adaptation from single-agent to multi-agent settings.	Improved research reproducibility and model interpretability through standardized pipelines.
Lower data acquisition costs, as all data is simulated.	Contribution to academic and industrial innovation in intelligent robotic control and multi-agent coordination.
Full compatibility between local environments for reproducible results.	Establish a baseline reference policy for comparison in future reinforcement learning research.

Next Steps

Since this project focused on manually simulating and testing multiple RL models, the next phase will **refine the Gazebo + ROS 2 environment and migrate the A2C baseline** into that setup. The goal is to maintain a lighter, computationally efficient architecture, given that the analyzed model originally involved around 31 features. Future work includes:

- Create a Jupyter Notebook to exclusively load metrics, generate the 4 graphs (for the metric evaluation), and export the figures for reports or LaTeX documents.
- Prepare the environment and documentation to facilitate the transition to the multi-agent framework.
- Refine the baseline with the characteristics previously defined (stronger penalties, exploitation implementation).

*In **coordination with the team**, the following activities will be also carried out:*

- *Defining success gates for comparing single-agent and multi-agent simulations.*
- *Reviewing and validating each iteration before approving the transition from the single-agent to the multi-agent setup.*
- *Establishing fixed maps, random seeds, and run budgets to ensure fair and consistent comparisons across experiments.*

Thank You!

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